औषध निरीक्षक, अन्न व औषधे प्रशासकीय सेवा, गट-ब
वैद्यकीय शिक्षण व औषधी द्रव्ये विभाग

Drugs Inspector, Food and Drugs Administrative Service, Gr.-B
Medical Education & Drugs Department

परीक्षेचे टप्पे : लेखी परीक्षा - २०० गुण **मुलाखत** - ५० गुण

-: परीक्षा योजना :-

विषय व संकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विहित अभ्यासक्रमावर आधारीत (११३२)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विहित अभ्यासक्रमावर आधारीत संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.		Topic
1)		**Physiology, Pathophysiology, Pharmacology and Toxicology:**
	1.1	**Physiology-** **i)** Whole human blood (I.P.) **ii)** Composition, function of Blood elements **iii)** Coagulation of blood, Blood groups **iv)** Blood components, manufacturing and storage **v)** Blood bank- operation, requirement and standards
	1.2	**Pathophysiology-** **i)** STD, AIDS, **ii)** Hypersensitivity, **iii)** Hepatitis and HBSAg.
	1.3	**Pharmacology –** Chemistry, SAR, Factors modifying, drug action, and mechanism of drugs belonging to following category- **i)** Coagulation and hematinics **ii)** Anti-hypertensives, CHF, Angina, arrhythmia, diuretics **iii)** CNS drugs – sedatives, hypnotics, local anesthetics, NSAIDS, opioids, analgesics, antipyretics, antitussive **iv)** Chemotherapeutic agents – anticancer, antifungal, antibiotics, anti-tubercular agents, urinary tract anti-infective, antiviral **v)** Anti-diabetics, contraceptives, endocrine and Hormones, uterine drugs **vi)** Respiratory drugs – anti-asthmatics, COPD, expectorants **vii)** GIT drugs – antiulcer, constipation, diarrhea, emetics & antiemetic, digestants, **viii)** Autocoids, Immuno-stimulants and immuno-suppressants **ix)** Introduction to drug design and combinatorial chemistry

	1.4	Toxicology: Classification of poison, chelating agents, heavy metal, insecticide poisoning, antidotes Sign, symptoms and management of CNS drug poisons. Genotoxicity, carcinogenicity, teratogenicity, mutagenicity
2)	**Microbiology :**	
	2.1	**i)** Sterilization principles and procedures, Aseptic techniques, Disinfections, pyrogen and pyrogen testing **ii)** Assay of antibiotics and vitamins **iii)** Biochemical serological, virological techniques, **iv)** BoD, different media **v)** Microbial drugs, drug resistance, microbial limit test, Pathogenic organism of common occurrence.
3)	**Biochemistry and Chemistry of drugs / Pharmaceuticals :**	
	3.1	Biochemistry: **i)** Metabolism of carbohydrates, proteins and lipids **ii)** DNA, RNA, nucleic acid and components, Enzymes **iii)** Recombinant technology and its applications **iv)** Immunological products – vaccines, sera
	3.2	Chemistry: Isomers, tautomers, fermentation, synthesis of antibiotics, diazo reaction, chemistry of steroids
	3.3	Pre-formulation studies:-Physical and chemical properties, Preservatives, suspending, emulsifying agents, different additives, vehicles, types of waters Pharmaceuticals: **i)** Dosage forms : Tablet, liquid orals, capsules, parenterals, opthalmics etc. **ii)** Novel drug delivery systems - sustained release / controlled drug delivery system, microencapsulation, mucosal, implantable, transdermal, gastroretentive, nasopulmonary, targated, ocular, intrauterine drug delivery system **iii)** Storage condition (raw materials and pharmaceuticals) **iv)** Medical devices. **v)** Route of administration Industrial Pharmacy **i)** Pilot plant scale up techniques **ii)** Technology development and transfer **iii)** Regulatory authorities and requirements Cosmetic science **i)** Formulation of cosmetics and their standards. **ii)** Herbal drug technology: **iii)** Herbs as a pharmaceutical raw material **iv)** Indian system of medicine – ayurveda, siddha, unani, homeopathy **v)** Herbal formulations like syrups, mixtures and tablets and Novel dosage forms like phytosomes **vi)** Neutraceuticals and herbal cosmetics
	3.4	Principal and application of analysis using **i)** UV, Atomic absorption & IR spectroscopy, NMR, Mass spectrometry **ii)** Thermal and x ray diffraction analysis **iii)** Chromatography : HPLC, GLC, HPTLC, LCMS **iv)** Hyphenated analytical techniques. Limit tests **v)** Radioimmunoassay Quality assurance and quality management system, Total quality management, ICH guidelines, Quality by Design, Documentation in pharmaceutical industry, calibration and validation of instrument
	3.5	Pharmacy Practice: **i)** Community pharmacy and management, wholesale store management, retail store management **ii)** Hospital pharmacy and its management **iii)** Drug distribution in hospital
4)	**Regulations related to drugs and General knowledge related to Pharmaceuticals**	
	4.1	**Drugs and cosmetic Act 1940 and rules 1945** **i)** Definitions- Drugs, Cosmetics, Medical Devices, **ii)** Statutory bodies- DTAB, DCC etc.

		iii) Standard quality, misbranded, adulterated, Spurious drugs and cosmetics. **iv)** Government analysts - Duties, procedures **v)** Drug inspector - Duties, power, procedures, **vi)** Import,Manufacture, sale and distribution of drugs. **vii)** Good Manufacturing Practices and Good Laboratory Practices. **viii)** Schedules under Drug and Cosmetics Rules. **ix)** Cosmetics Rules 2020 **x)** Medical Devices Rules 2017
	4.2	**i)** Drugs and Magic Remedies and Objectionable Advertisement Act 1954 and Rules **ii)** Narcotic Drugs and Psychotropic Substances Act 1985. Essential narcotic drugs, RMI, Maharashtra NDPS Rules. **iii)** Drugs Price Control Order 2013 (DPCO) and NPPA, Essential Narcotic drugs, **iv)** Poisons Act 1919 and Maharashtra poison rules 1970 **v)** The Pharmacy Act 1948 **vi)** Intellectual property rights (IPR) **vii)** Consumer protection Act **viii)** Right to information Act. **ix)** Neutraceuticals. **x)** WHO-GMP certification **xi)** National health program **xii)** Irrational drug combinations **xiii)** Antimicrobial resistance a global threat **xiv)** Indian Pharmacopoeia Commission (IPC) **xv)** Adverse Drug Reactions (ADR)
5)	**Intelligence Test:**	
	Questions will be asked to test how far candidates can think quickly and accurately.	

Note:- Wherever Acts/Rule/Regulations are mentioned in syllabus, latest amendments/versions to be considered .

दिनांक – २९ सप्टेंबर, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग